

Thread Test Assignment

Ishtiaq-Thread.java

```
import java.util.concurrent.atomic.AtomicLong;

public class Ishtiaq-Thread implements Runnable {

    static AtomicLong safeCount = new AtomicLong(0);
    static long unsafeCount = 0;

    long localCount = 0;
    final long increments;
    final boolean safe;

    Ishtiaq-Thread(long increments, boolean safe) {
        this.increments = increments;
        this.safe = safe;
    }

    @Override
    public void run() {
        for (long i = 0; i < increments; i++) {
            if (safe) safeCount.incrementAndGet();
            else unsafeCount++;
            localCount++;
        }
    }
}
```

```

public static void main(String[] args) throws
    InterruptedException {
    int n = Integer.parseInt(args[0]);
    long m = Long.parseLong(args[1]);
    boolean safe = Boolean.parseBoolean(args[2]);

    Ishtiaq_Thread[] tasks = new Ishtiaq_Thread[n];
    Thread[] threads = new Thread[n];
    for (int i = 0; i < n; i++) {
        tasks[i] = new Ishtiaq_Thread(m, safe);
        threads[i] = new Thread(tasks[i]);
        threads[i].start();
    }

    for (Thread t: threads) t.join();

    long nonStatic = 0;
    for (Ishtiaq_Thread t: tasks) nonStatic +=
        t.localCount;

    long staticCount = safe ? safeCount.get() :
        unsafeCount;

```

long diff = Math.abs(StaticCount - nonStaticCount);

String pct;

if (nonStatic == 0) pct = StaticCount == 0 ? "0.00":

"undefined";

else pct = String.format("%.4f", diff * 100.0 / nonStatic);

System.out.println("Threads" + n);

System.out.println("Increments" + m);

System.out.println("Mode" + (safe ? "safe":
"unsafe));

System.out.println("Expected" + (n * m));

System.out.println("Static Count: " + StaticCount);

System.out.println("Non-static Count" + nonStatic);

System.out.println("Absolute difference" + diff);

System.out.println("Difference(%.):" + pct);

}

Compile and Run:

```
javac Ishtiak-Thread.java
```

```
java Ishtiak-Thread 10 50000 true
```

```
java Ishtiak-Thread 10 50000 false
```

Output:

① Java Ishtiak-Thread 10 50000 true

Threads : 10

Increments : 50000

Mode : safe

Expected : 500000

Static count : 500000

non-static count : 500000

Absolute difference : 0

Difference (%) = 0.00

② Java Ishtiak-Thread 10 50000 false

Threads : 10

Increment : 50000

Mode : unsafe

Expected : 500000

Static count : 68687

Non static count : 500000

Absolute difference : 431313

Difference (%) : 86.2626